WORKED SOLUTION

STEP II 2005, Question 3

Pure • Integration by Parts

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q3

The key idea

Differentiate first: the derivative of $\sin x - x\cos x$ collapses to something with an obvious sign, and that gives the sketch and the bounds on y in one go. The two integrals are routine integration by parts, helped by double-angle formulae for the squared terms. The final inequality comes from comparing them: a number between 0 and 1 is at least its own square.

Solution

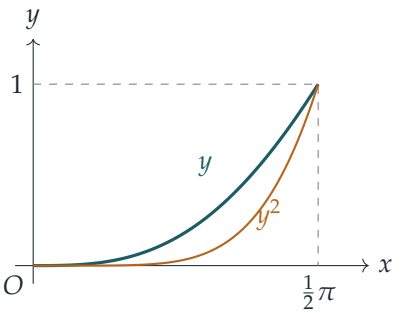
The sketch and the bounds

Differentiating with the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos x - (\cos x - x\sin x) = x\sin x.$$

For $0 < x \leq \frac{1}{2}\pi$ both x and $\sin x$ are positive, so y is strictly increasing on $[0, \frac{1}{2}\pi]$. Its least value is therefore $y(0) = 0 - 0 = 0$ and its greatest is $y(\frac{1}{2}\pi) = 1 - \frac{1}{2}\pi \cdot 0 = 1$. Hence $0 \leq y \leq 1$, and in fact $0 < y < 1$ for $0 < x < \frac{1}{2}\pi$.

For the shape, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \sin x + x\cos x > 0$ on $(0, \frac{1}{2}\pi]$, so the curve bends upwards throughout. At the origin the gradient is 0. Using the Maclaurin series, $y = (x - \frac{1}{6}x^3) - x(1 - \frac{1}{2}x^2) + \cdots = \frac{1}{3}x^3 + \cdots$, so the curve leaves the origin very flatly, like a cubic. At $x = \frac{1}{2}\pi$ the gradient is $\frac{1}{2}\pi$.



The amber curve y^2 is shown too, since it is the key to the last part.

Part (i)

We need $\int_0^{\pi/2} x\cos x\,\mathrm{d}x$. Integrating by parts with $u = x$, $\mathrm{d}v = \cos x\,\mathrm{d}x$,

$$\int_0^{\pi/2} x\cos x\,\mathrm{d}x = \Big[x\sin x\Big]_0^{\pi/2} - \int_0^{\pi/2} \sin x\,\mathrm{d}x = \tfrac{1}{2}\pi - 1.$$

Since $\int_0^{\pi/2} \sin x\,\mathrm{d}x = 1$,

$$\int_0^{\pi/2} y\,\mathrm{d}x = 1 - (\tfrac{1}{2}\pi - 1) = 2 - \frac{\pi}{2},$$

as required.

Part (ii)

Expanding,

$$y^2 = \sin^2 x - 2x\sin x\cos x + x^2\cos^2 x = \sin^2 x - x\sin 2x + \tfrac{1}{2}x^2 + \tfrac{1}{2}x^2\cos 2x,$$

using $2\sin x\cos x = \sin 2x$ and $\cos^2 x = \frac{1}{2}(1+\cos 2x)$. We take the four pieces in turn. First, $\sin^2 x = \frac{1}{2}(1-\cos 2x)$, and $\sin 2x$ vanishes at both limits, so

$$\int_0^{\pi/2} \sin^2 x\,\mathrm{d}x = \Big[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\Big]_0^{\pi/2} = \frac{\pi}{4}.$$

Next, by parts with $u = x$, $\mathrm{d}v = \sin 2x\,\mathrm{d}x$, and using $\cos\pi = -1$,

$$\int_0^{\pi/2} x\sin 2x\,\mathrm{d}x = \Big[-\tfrac{1}{2}x\cos 2x\Big]_0^{\pi/2} + \tfrac{1}{2}\int_0^{\pi/2} \cos 2x\,\mathrm{d}x = \frac{\pi}{4} + 0 = \frac{\pi}{4}.$$

Third, $\int_0^{\pi/2} \frac{1}{2}x^2\,\mathrm{d}x = \frac{1}{6}(\frac{1}{2}\pi)^3 = \dfrac{\pi^3}{48}$.

Finally, by parts with $u = x^2$, $\mathrm{d}v = \cos 2x\,\mathrm{d}x$; the boundary term $[\frac{1}{2}x^2\sin 2x]_0^{\pi/2}$ is zero, and the integral left over is the one we have just found:

$$\int_0^{\pi/2} x^2\cos 2x\,\mathrm{d}x = 0 - \int_0^{\pi/2} x\sin 2x\,\mathrm{d}x = -\frac{\pi}{4},$$

so the last piece contributes $\frac{1}{2} \cdot (-\frac{1}{4}\pi) = -\frac{1}{8}\pi$.

Adding the four pieces,

$$\int_0^{\pi/2} y^2 \, \mathrm{d}x = \frac{\pi}{4} - \frac{\pi}{4} + \frac{\pi^3}{48} - \frac{\pi}{8} = \frac{\pi^3}{48} - \frac{\pi}{8},$$

as required.

The deduction

? **Ask yourself** How can two integrals give an inequality about π?

Compare the integrands. Since $0 \le y \le 1$ we have $y - y^2 = y(1-y) \ge 0$ on the whole interval. Better, for $0 < x < \frac{1}{2}\pi$ we showed $0 < y < 1$, so $y(1-y) > 0$ there. A continuous function that is non-negative, and strictly positive on an open interval, has a strictly positive integral. Therefore

$$\int_0^{\pi/2} y^2 \, \mathrm{d}x < \int_0^{\pi/2} y \, \mathrm{d}x, \qquad \text{that is,} \qquad \frac{\pi^3}{48} - \frac{\pi}{8} < 2 - \frac{\pi}{2}.$$

Multiplying by 48 (positive, so the inequality is preserved),

$$\pi^3 - 6\pi < 96 - 24\pi, \qquad \text{so} \qquad \pi^3 + 18\pi < 96.$$

This is the required result.

Checks

- Numerically, $2 - \frac{1}{2}\pi \approx 0.4292$ and $\frac{1}{48}\pi^3 - \frac{1}{8}\pi \approx 0.2533$; Simpson's rule on the original integrands gives the same values to twelve decimal places. Both lie between 0 and $\frac{1}{2}\pi \approx 1.571$, as they must since $0 \le y \le 1$.
- The second integral is smaller than the first, as the picture of y^2 under y suggests.
- With $\pi \approx 3.1416$, $\pi^3 + 18\pi \approx 31.006 + 56.549 = 87.55$, comfortably below 96.

Common slips

- Getting the sign of the boundary term wrong in $\int x \sin 2x \, \mathrm{d}x$: at $x = \frac{1}{2}\pi$, $\cos 2x = \cos \pi = -1$, so $-\frac{1}{2}x \cos 2x$ is $+\frac{1}{4}\pi$.
- Only claiming $\le$ in the final step. To get the strict inequality you need $y(1-y) > 0$ somewhere, not just ≥ 0.
- Forgetting to say why $y \le 1$: knowing $y(\frac{1}{2}\pi) = 1$ is not enough without the fact that y is increasing.

Beyond STEP

The final step is a small instance of a big idea: inequalities between functions integrate to inequalities between numbers, and a clever choice of function can trap a constant like π. The same strategy gives the famous proof that $\frac{22}{7} > \pi$ from $\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x > 0$, and, more seriously, Niven's proof that π is irrational. In a first analysis course you will prove carefully that a continuous, non-negative function which is positive somewhere has a positive integral. Abbott's *Understanding Analysis* is a friendly place to see this done properly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.